A General Framework for Low Soundness Homomorphism Testing

ITCS 2026



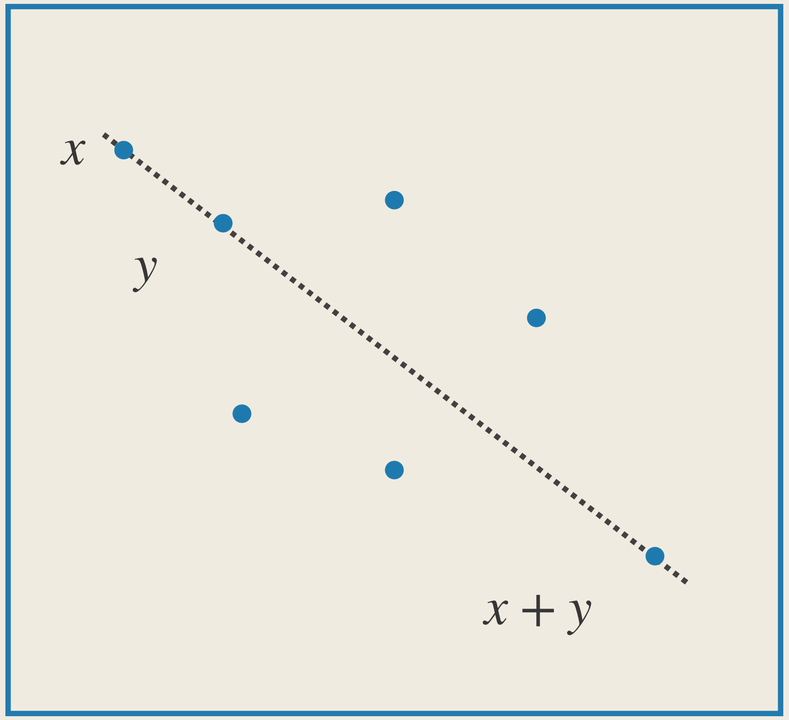
TUSHANT MITTAL
STANFORD UNIVERSITY

Presented by
Keller Blackwell



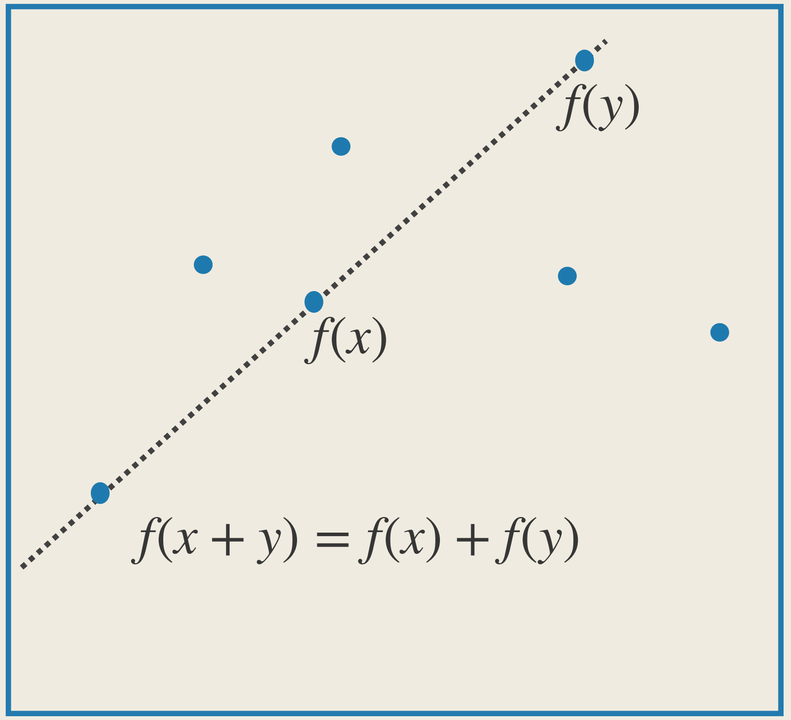
SOURYA ROY
UNIVERSITY OF IOWA

Linear functions



f : a linear function

“Preserves linearity”



Group Homomorphisms

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Group

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Set & “group operation” (★)

Group Homomorphisms

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Set & “group operation” ($\star$)

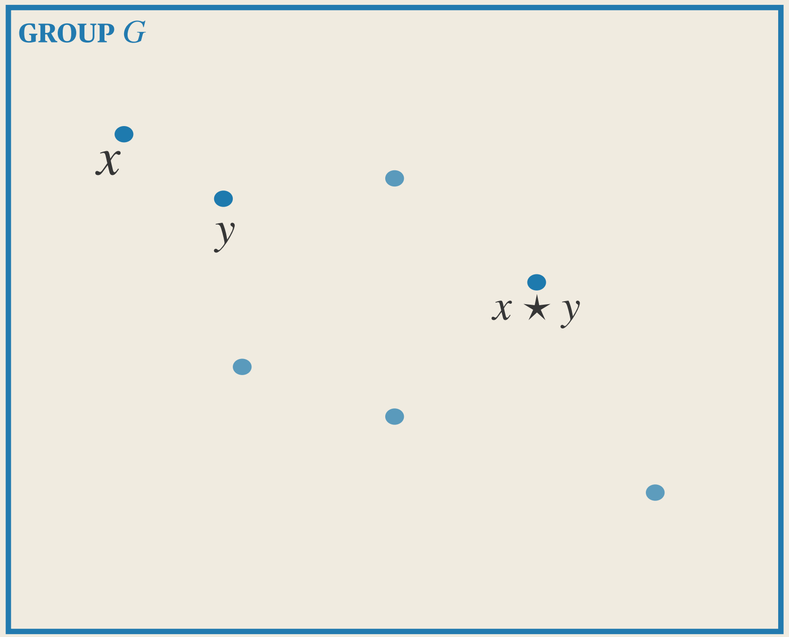
eg. $\mathbb{Z}_n = \{0,1,\ldots,n-1\}$ & addition modulo n

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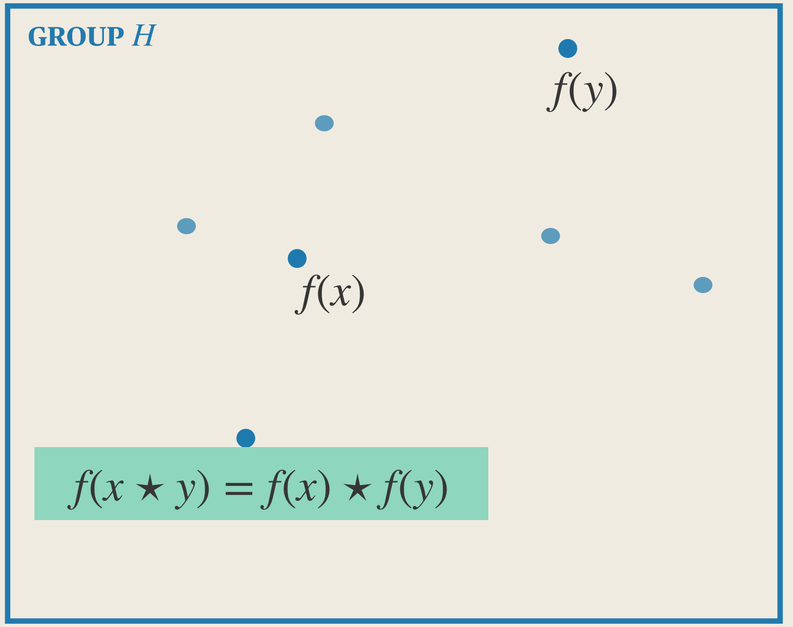
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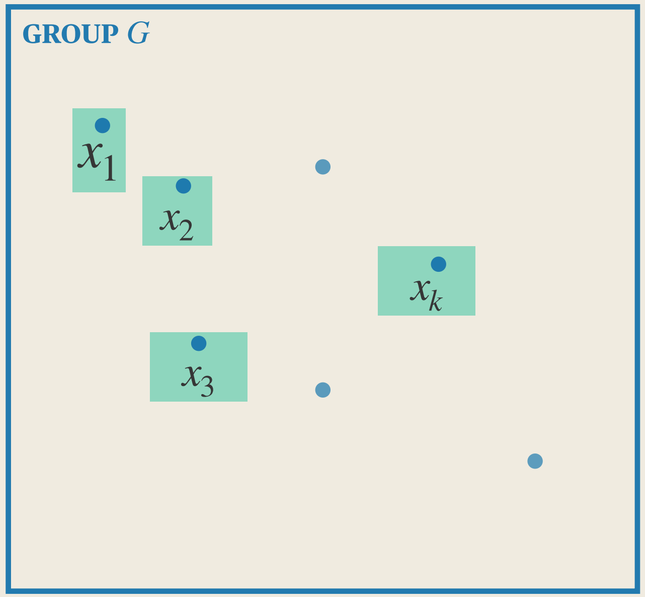


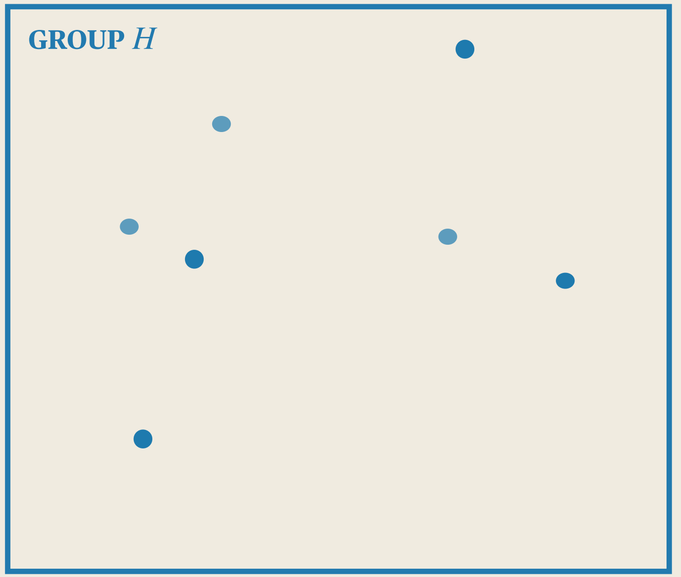
f : Homomorphism

"Preserves group operation"

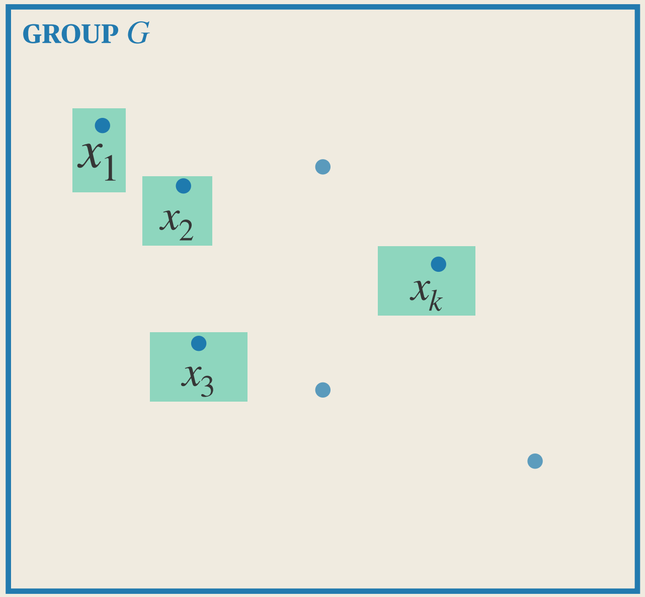


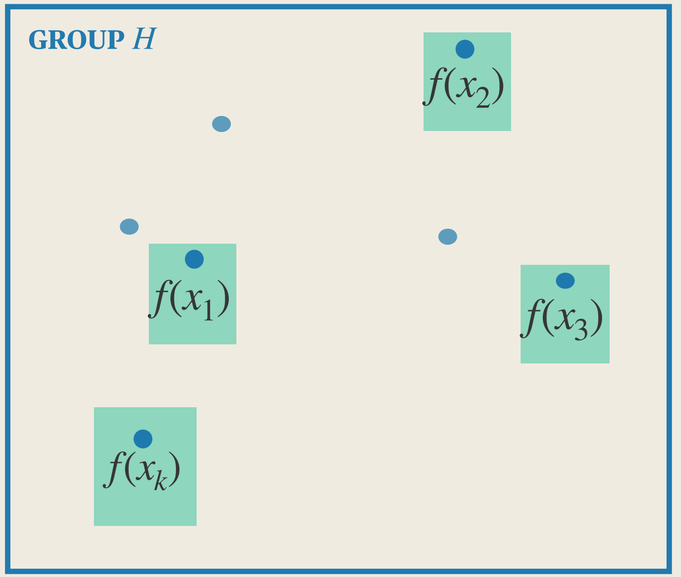
Homomorphism Test



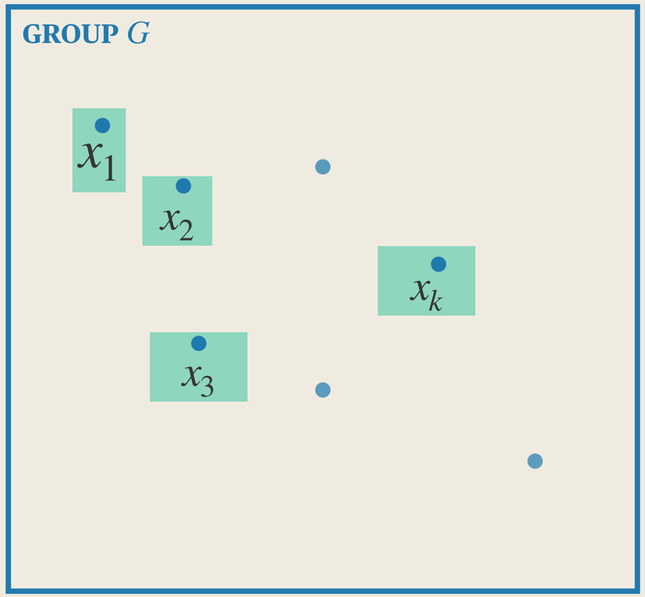


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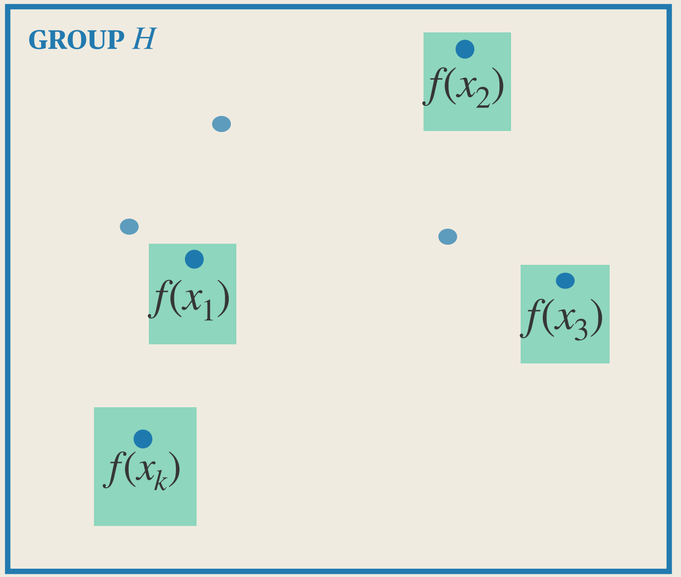


Homomorphism Test



A k-query test queries $f(x_1), \cdots, f(x_k)$

Outputs Yes/No.



“Probabilistically

Checkable

Proof (PCP)”

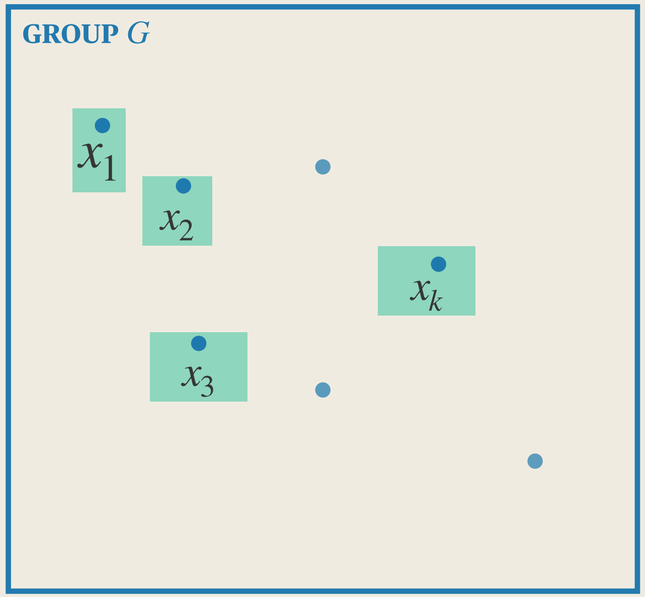


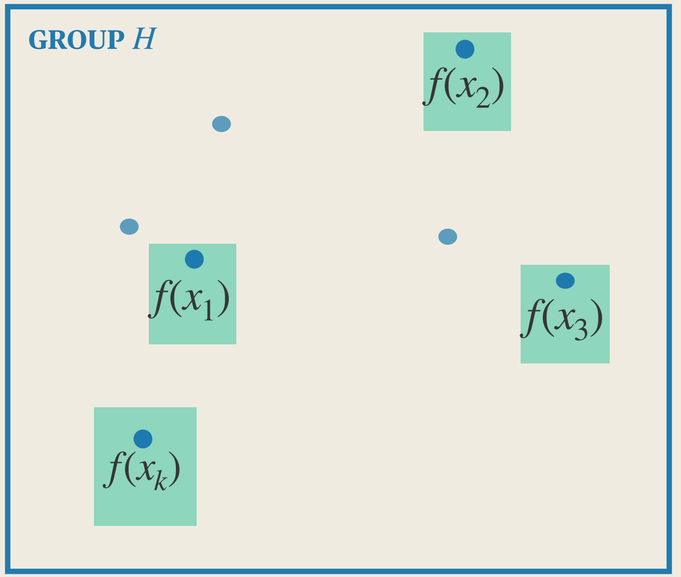
Kristina Armitage/ Quanta Magazine

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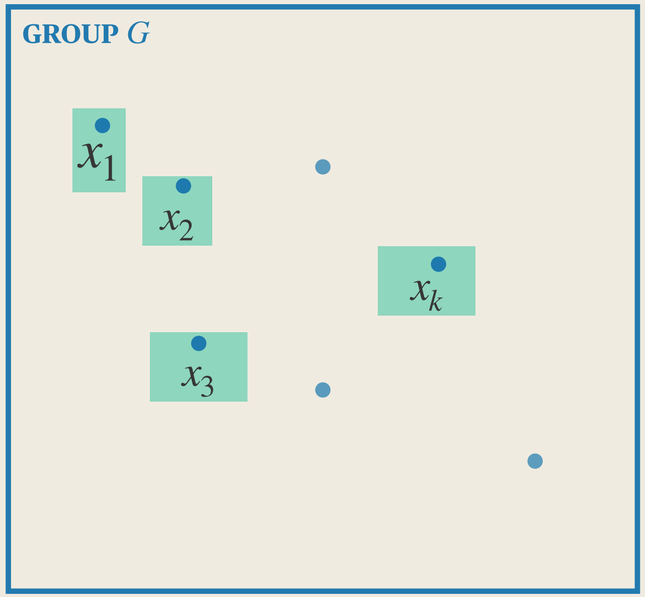
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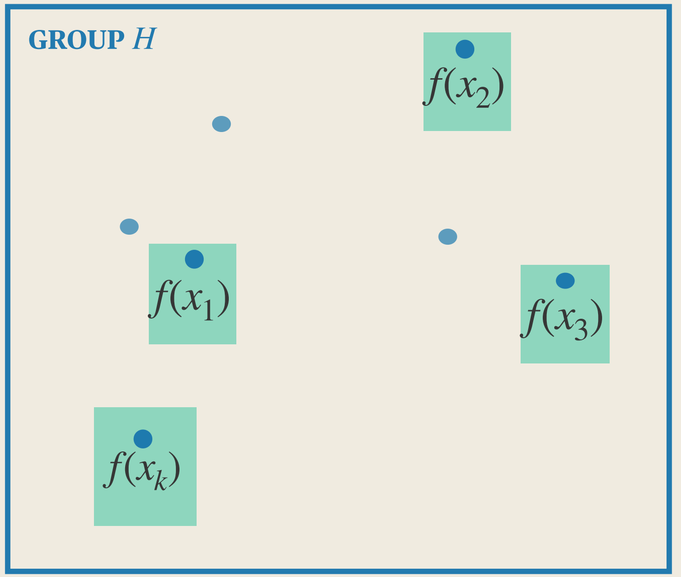
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Completeness

If f is a homomorphism,

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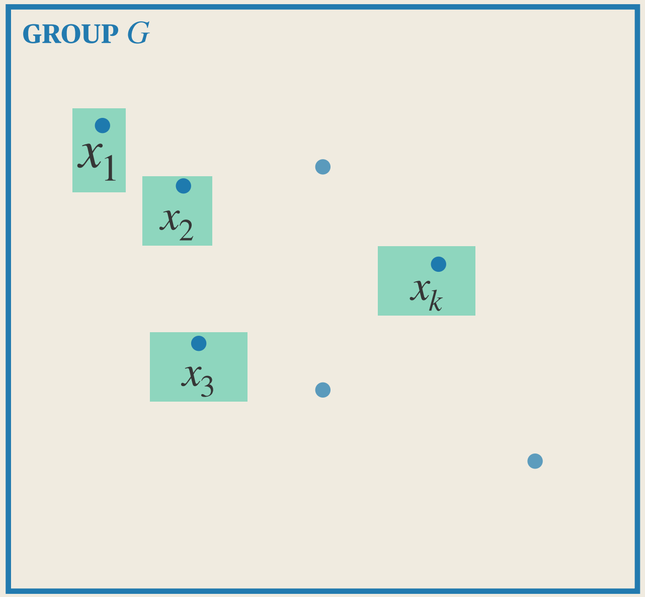




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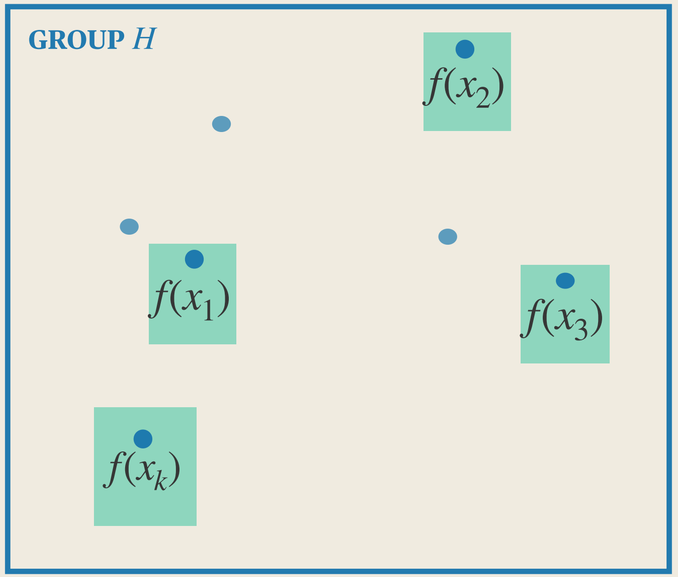
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Soundness

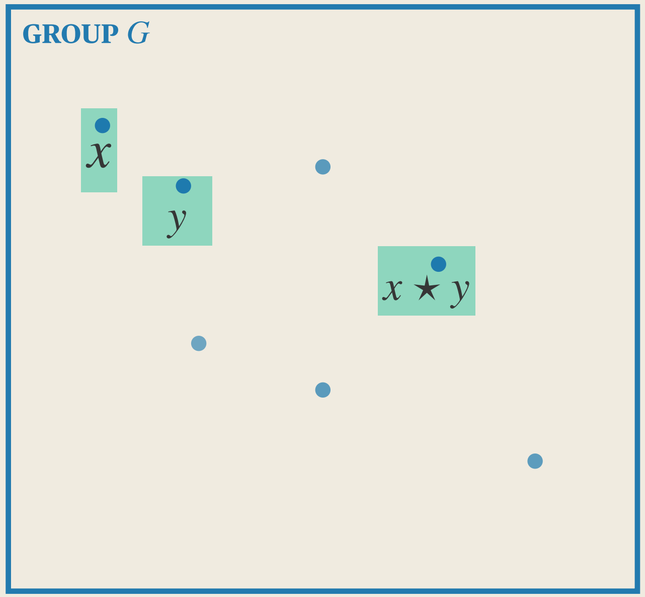
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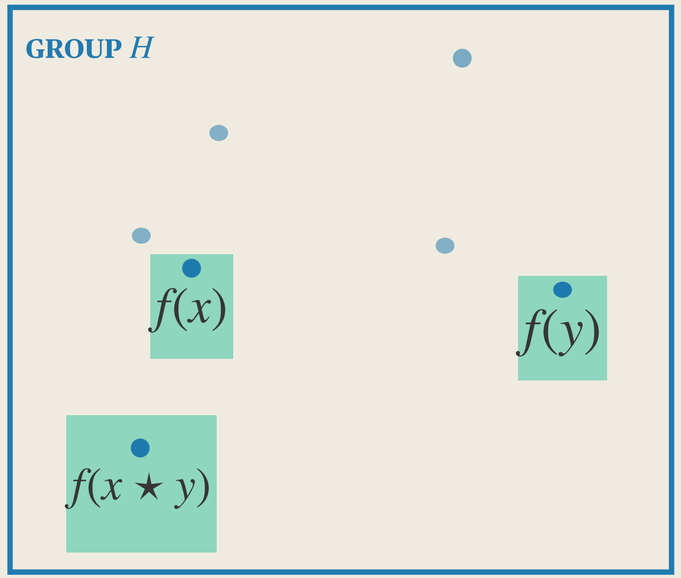
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$$\mathrm{agr}(\varphi, f) \;:=\; \Pr_{x \sim G}[f(x) = \varphi(x)] \;\geq\; \mathrm{s}(\delta)\,.$$



BLR Test: The only known general test

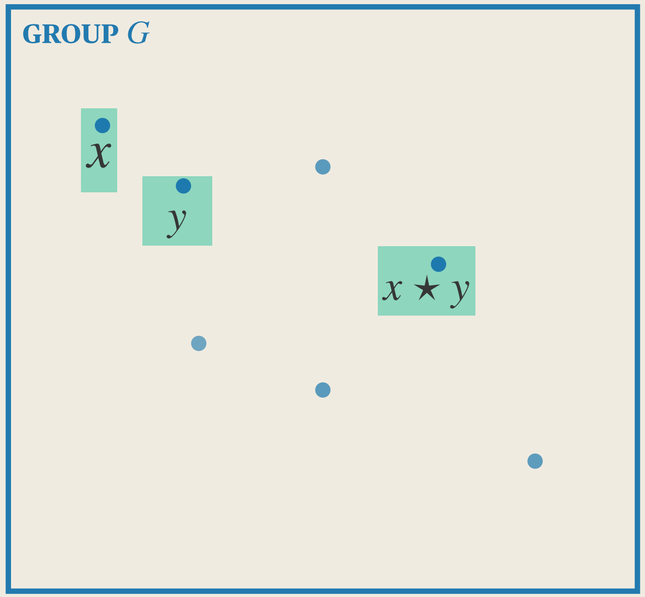


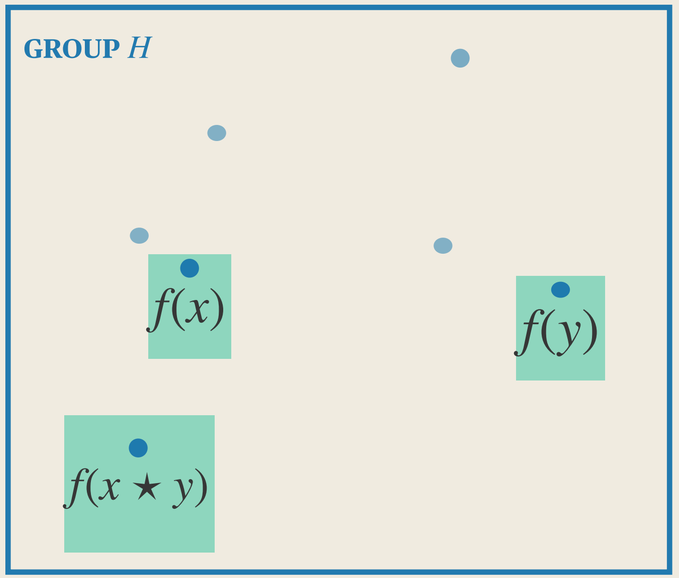


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[Blum–Luby–Rubinfeld'90] : One random pair (x, y)

Outputs Yes iff $f(x) \star f(y) = f(x \star y)$.

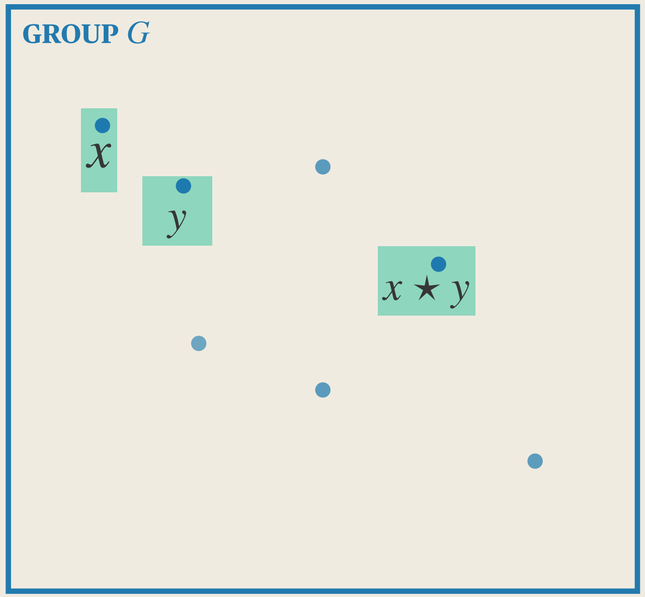




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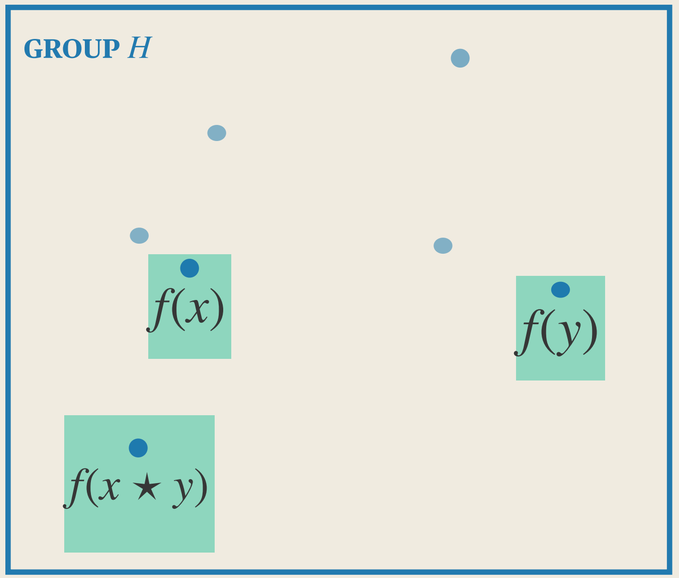
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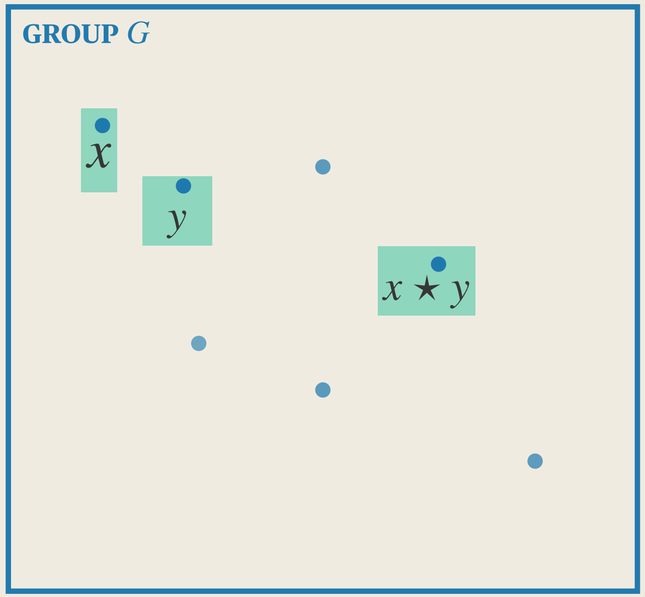
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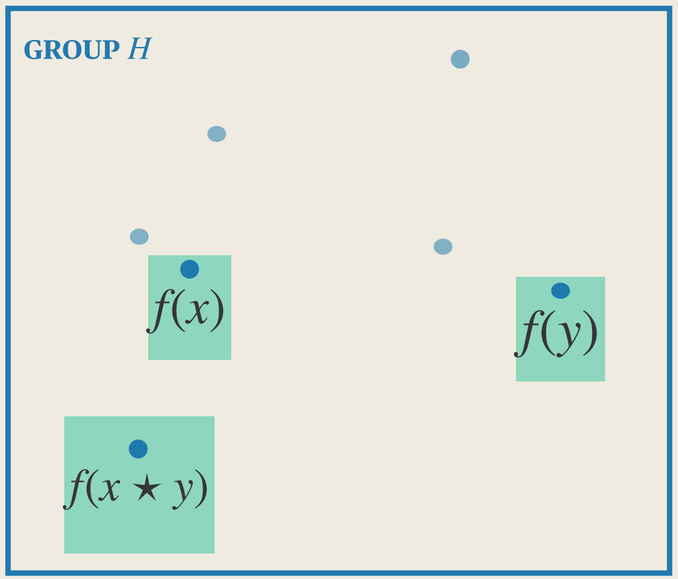
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BLR poorly understood in the low soundness regime

HIGH SOUNDNESS/UNIQUE DECODING

$\left(\delta > \frac{7}{9}\right)$

LOW SOUNDNESS/LIST DECODING

(ANY NON-TRIVIAL δ)

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$f : G \to H$

Any finite G, H

[Blum-Luby-Rubinfeld' 90,
Ben-Or, Coppersmith,Luby, Rubinfeld '04,
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LOW SOUNDNESS/LIST DECODING (ANY NON-TRIVIAL δ)

Citation	Setting	Agreement
[Bellare-Coppersmith-Hästad-Kiwi Sudan'95, Hästad-Wigderson'02]	$\mathbb{Z}_p^n \to \mathbb{Z}_p$	$\frac{\delta}{p}$
[Samrodsnitsky'07, Sanders'12, Gowers-Green-Manners-Tao'23]	$\mathbb{Z}_2^n \to \mathbb{Z}_2^m$	δ^9

Improved to δ by [Kiwi'03]

Same as Freiman-Rusza Problem

BLR generally does not work in low soundness!

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[Coppersmith'89]

BLR fails for cyclic groups in low soundness

BLR generally does not work in low soundness!

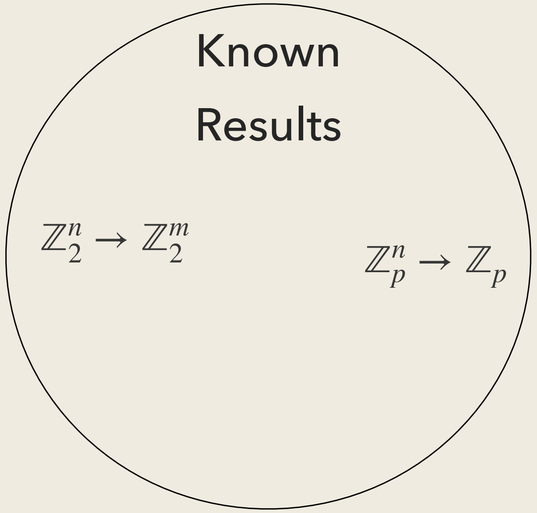
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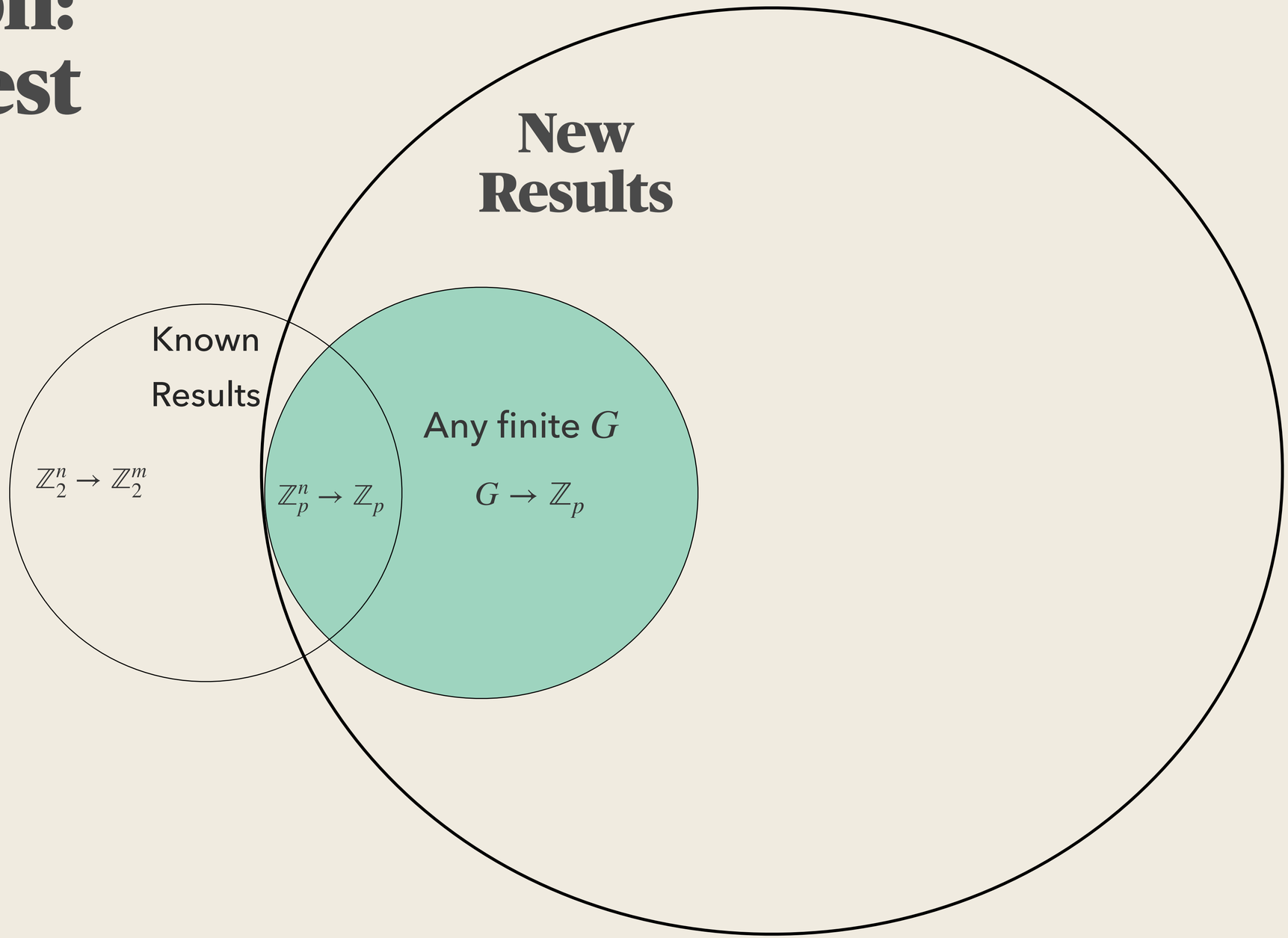
Key Question: Tests for a general pair of groups?

Our Contribution:
A new general test

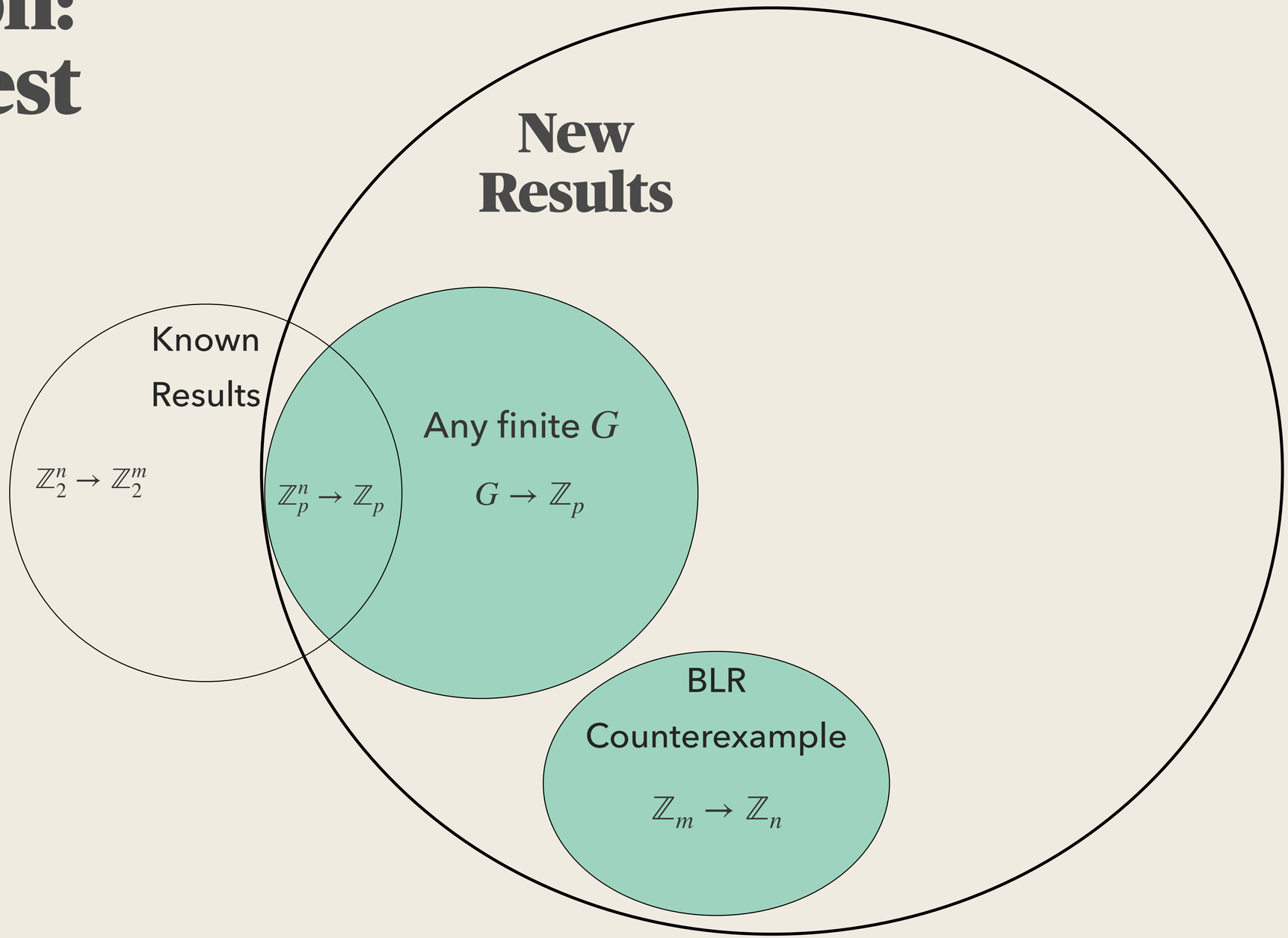
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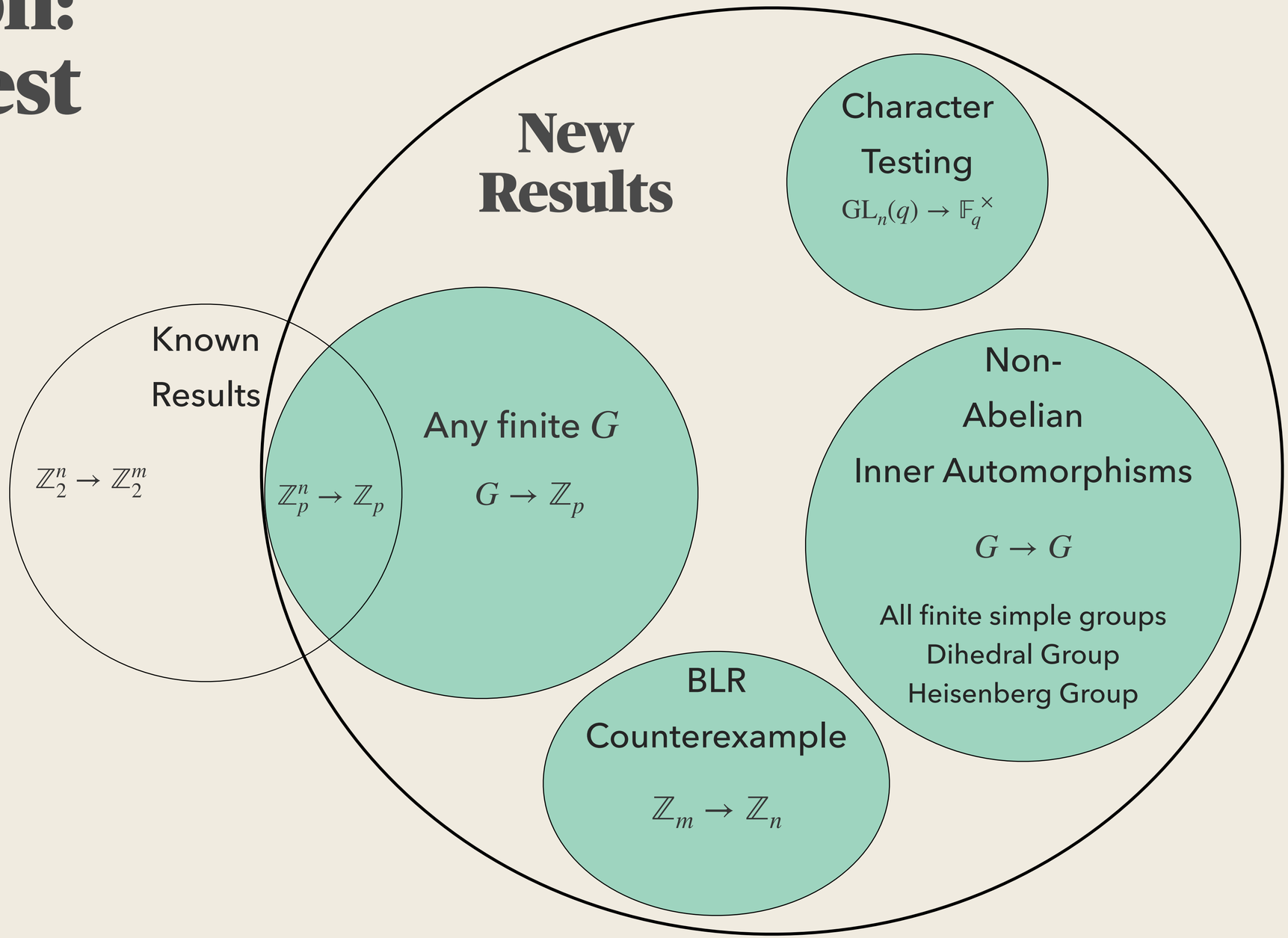
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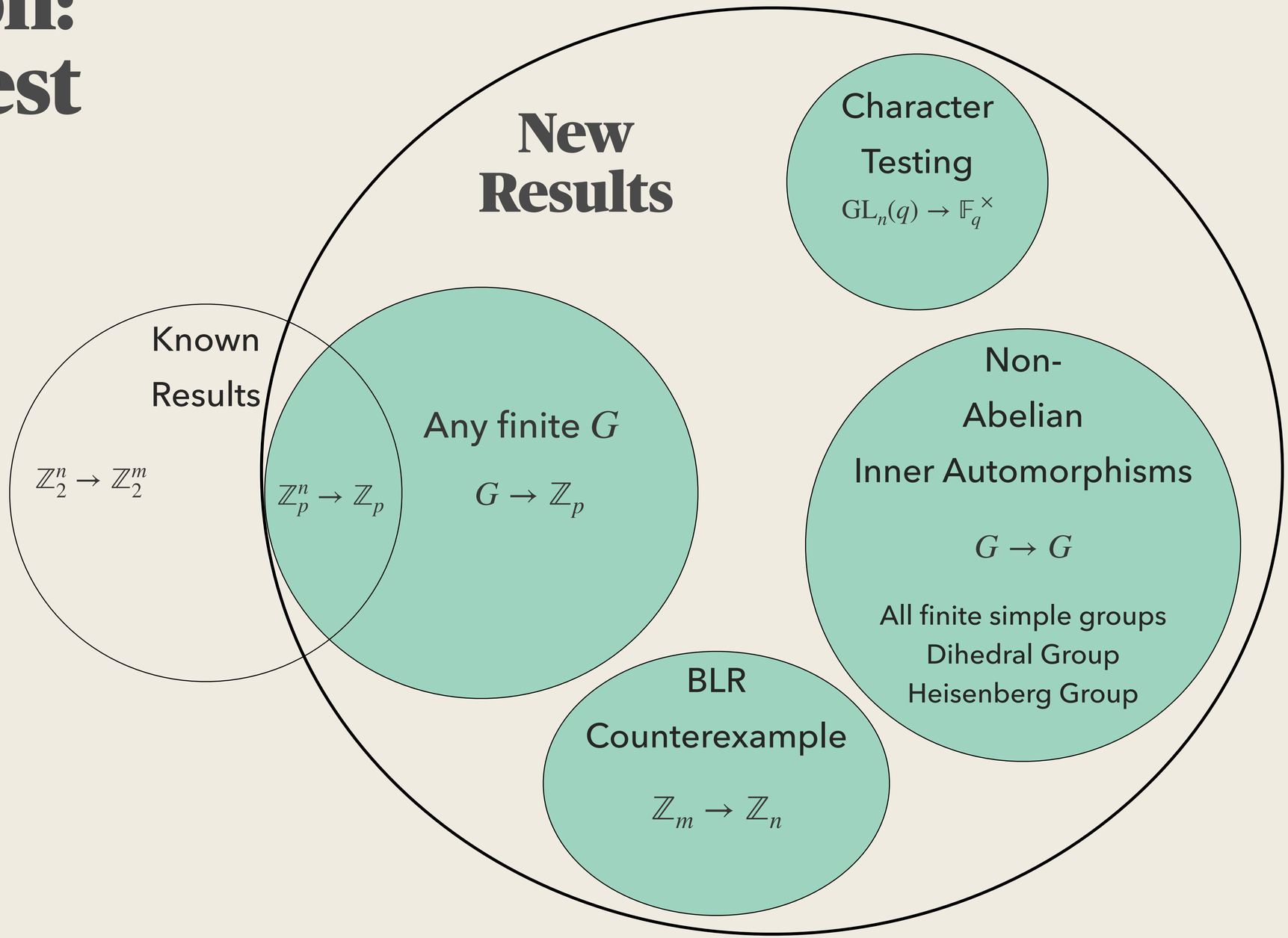


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Main New Idea – Queries not uniform (BLR) but a special *group-specific* distribution



Feature 1: Query vs Soundness

	G	H	Queries (any $k \geq 1$)	Soundness $[s(\delta)]$
Recovers the known results [BLR, Kiwi'03]	$\mathbb{Z}_q^n$	$\mathbb{Z}_q$	Odd, $2+k$	$O(\delta)^{\frac{1}{k}}$
Overcomes Coppersmith's counterexample	$\mathbb{Z}_{p^r}$	$\mathbb{Z}_{p^s}$	$2+k$	
	$\mathbb{Z}_{p^r}$	Abelian group of p-rank $\leq t$	$t+1+k$	
	$\mathbb{Z}_n$	$\mathbb{Z}_m$	$3+k$	
	$\mathbb{Z}_q$	$\mathbb{Z}_q^n$	$1+k$	

Feature 2: List Decoding Bounds

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Our Framework

k query test $\implies$

Number of homomorphisms with ϵ-agreement $\leq$ $O(\varepsilon^{1-k})$.

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G, H cyclic groups

(prime power order)

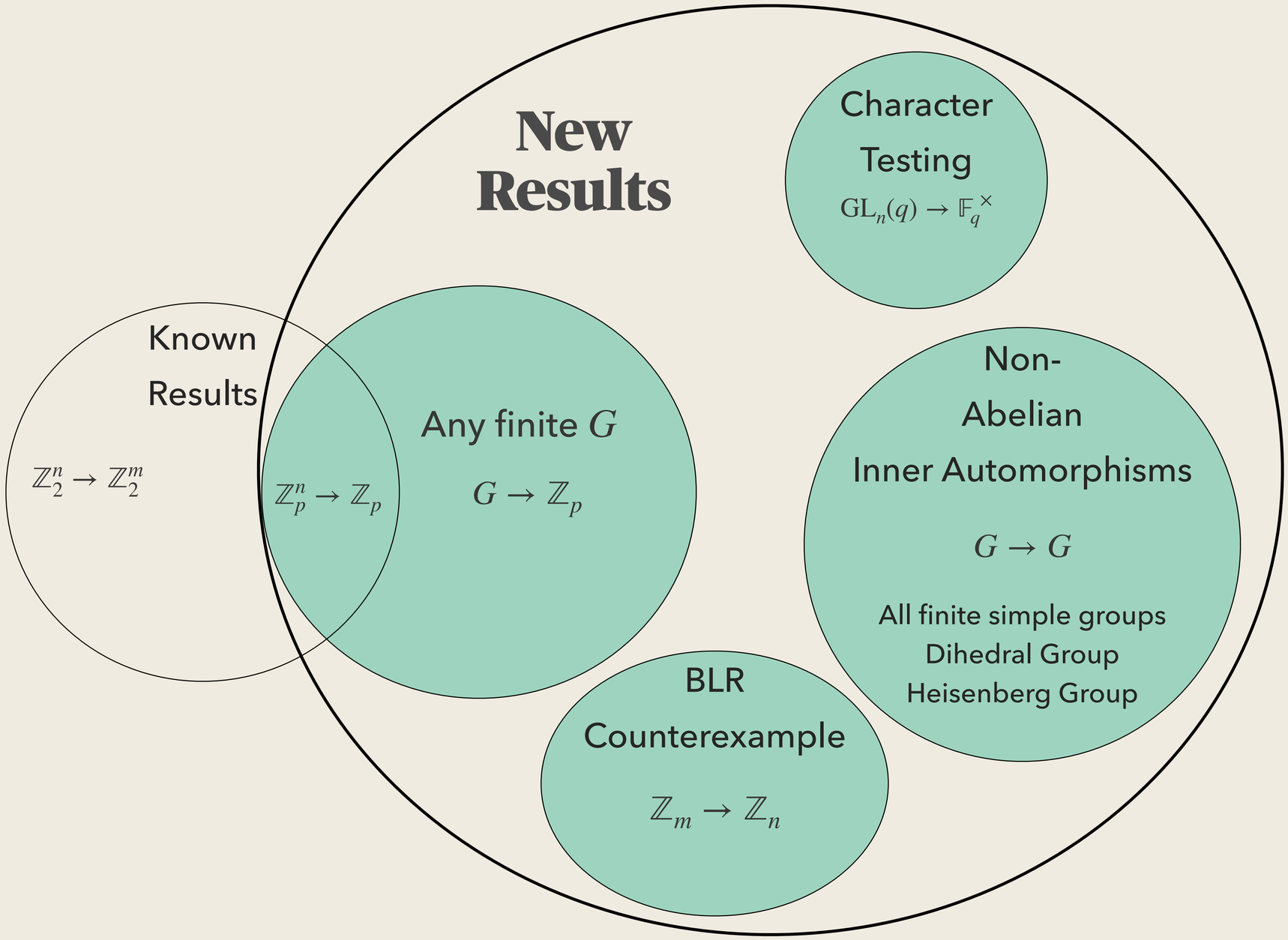
$|L_\varepsilon| \leq O(\varepsilon^{-105})$ → $|L_\varepsilon| \leq 2\varepsilon^{-2}$

[Dinur, Grigorescu, Kopparty, Sudan'08]

[Our Framework]

A New General Framework

- General but Group-Specific Tests
- Flexible Query vs Soundness
- List Decoding Bounds
- “Lifting” Property
- Fourier-free Proof



Thank You!